

**A REPORT
ON
INTRODUCTION TO ARTIFICIAL INTELLIGENCE
(BT104HS)**

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This report is submitted as part of the partial fulfilment of the requirements for B.Tech in Artificial Intelligence, 1st Year 1st Semester, 2026. Preparing this report was a valuable learning experience that helped deepen my understanding of the fundamental concepts of Artificial Intelligence — including its definition, goals, history, intelligent agents, the PEAS framework, search algorithms, and problem-solving methods.

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"INTRODUCTION TO ARTIFICIAL INTELLIGENCE"

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Abstract

This report provides a comprehensive study of the fundamental concepts of Artificial Intelligence (AI) as taught in BT104HS: Introduction to Artificial Intelligence. It covers seven core topics drawn directly from course notes and exam questions: the definition and goals of AI, the Turing Test, intelligent agents and their types, the PEAS framework for characterising agents, problem-solving techniques in AI, search algorithms (both uninformed and informed), and the history of AI including key milestones and the AI winter.

Artificial Intelligence is defined as the branch of computer science concerned with building machines that can learn, reason, and solve problems in ways that mimic human intelligence. The Turing Test (1950), proposed by Alan Turing, remains the most famous benchmark for machine intelligence. Intelligent agents — the core entities in AI — perceive their environment through sensors and act through actuators. They are categorised into five types: simple reflex, model-based, goal-based, utility-based, and learning agents. The PEAS framework (Performance Measure, Environment, Actuators, Sensors) provides a structured way to describe any intelligent agent system.

Problem-solving in AI relies on search algorithms. Uninformed algorithms such as Breadth-First Search (BFS) explore the search space without domain knowledge, while informed algorithms such as A* Search use heuristic functions to find solutions more efficiently. The history of AI spans from the 1940s to the present day and includes two major AI winters, followed by a modern revival driven by deep learning and large-scale computing. All content is drawn directly from handwritten course notes for BT104HS at Aryan School of Engineering, Kathmandu.

Chapter 1

Introduction to Artificial Intelligence

1.1 Definition of Artificial Intelligence

Figure 1: What is Artificial Intelligence? — Core Areas of Study

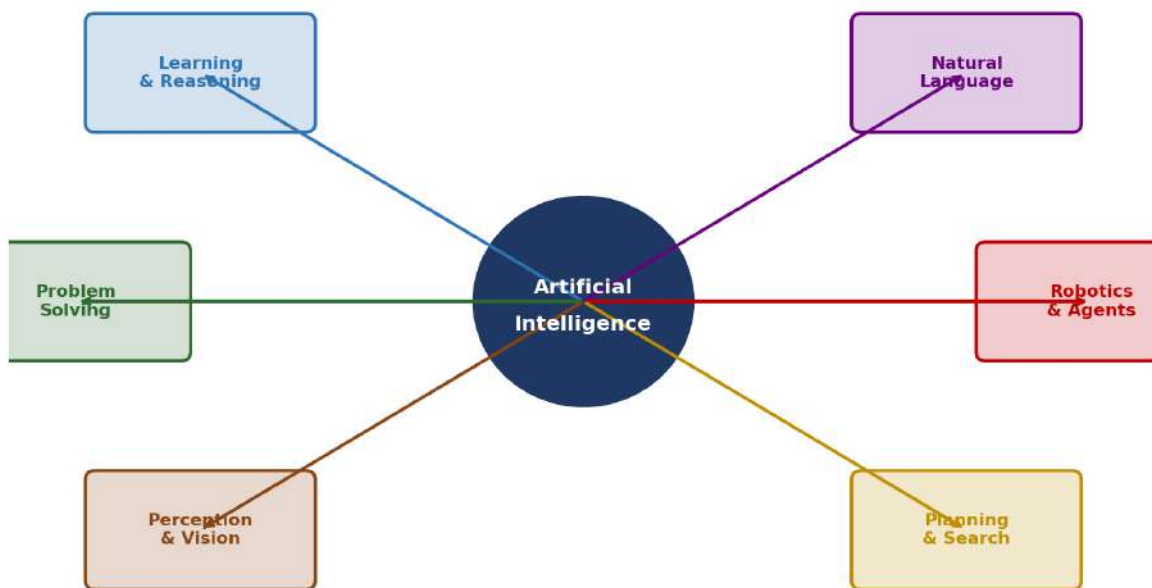


Figure 1: What is Artificial Intelligence? — Core Areas of Study

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think like humans and mimic their actions. It is the branch of computer science concerned with making computers behave like humans. AI is also defined as the science and engineering of making intelligent machines, especially intelligent computer programs.

AI can be applied to any machine that exhibits traits associated with the human mind such as learning and problem-solving. AI systems aim to understand human-like reasoning and decision making, working and reacting as humans would in similar situations.

John McCarthy, who coined the term, described AI as: "the science and engineering of making intelligent machines." AI is also known as Machine Intelligence — the intelligence

demonstrated by machines as opposed to the natural intelligence displayed by humans and animals.

1.2 Main Goals of Artificial Intelligence

The overall goal of AI is to develop technology that allows computers and machines to function in an intelligent manner. The following are the key goals of Artificial Intelligence:

- **Create Expert Systems:** To build systems that exhibit intelligent behaviour, learn from experience, demonstrate, explain, and advise users. These systems handle complex tasks that would normally require human expertise.
- **Augment Human Intelligence in Machines:** To replicate human mental capabilities in machines — enabling them to read, find, search, and process data intelligently. The aim is for machines to understand human language and thought processes.
- **Enable Natural Language Processing:** To develop systems that interact with humans naturally through spoken and written language, requiring understanding of context, grammar, and meaning.
- **Machine Perception:** To develop machines that use input from sensors (cameras, microphones, radar) to read and react to the physical environment around them.
- **Automate Reasoning and Problem-Solving:** To enable machines to reason logically, plan sequences of actions, and solve complex real-world problems with accuracy and efficiency.
- **Continuous Learning:** AI systems should improve their performance over time by learning from new data and experiences without being explicitly reprogrammed for every new situation.

Figure 2: The Six Main Goals of Artificial Intelligence

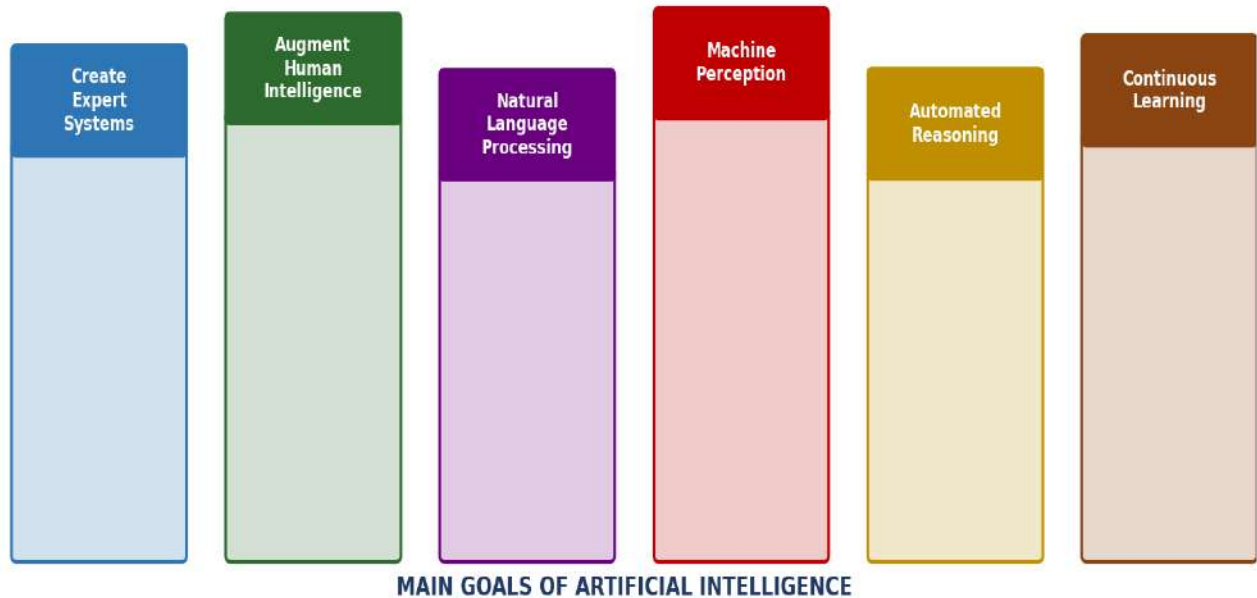


Figure 2: The Six Main Goals of Artificial Intelligence

1.3 Real-Life Examples of AI

Example 1 — AI in Healthcare: AI is used to build sophisticated machines that can detect diseases and identify cancer cells. AI systems analyse patient conditions using lab data and medical history to ensure early diagnosis. It combines historical data and medical intelligence for more accurate diagnostics, reducing human error and improving patient outcomes.

Example 2 — AI in Navigation and Self-Driving Cars: AI provides users with accurate, timely route information to improve safety. GPS systems use AI to improve efficiency. Self-driving vehicles use a combination of Convolutional Neural Networks (CNN) and Graph Neural Networks (GNN) to navigate roads, detect lanes, recognise obstacles, and make real-time driving decisions without human input.

Chapter 2

The Turing Test

2.1 Background and Origin

The Turing Test was introduced by British mathematician and scientist Alan Turing in 1950. He proposed it as a way to determine whether a computer machine can think intelligently like a human. Turing published his landmark paper "Computing Machinery and Intelligence" asking the fundamental question: "Can machines think?" A computer can be said to be an artificial intelligence if it can mimic human responses under specific conditions.

2.2 How the Turing Test Works

Figure 3: The Turing Test — Three-Player Imitation Game Setup

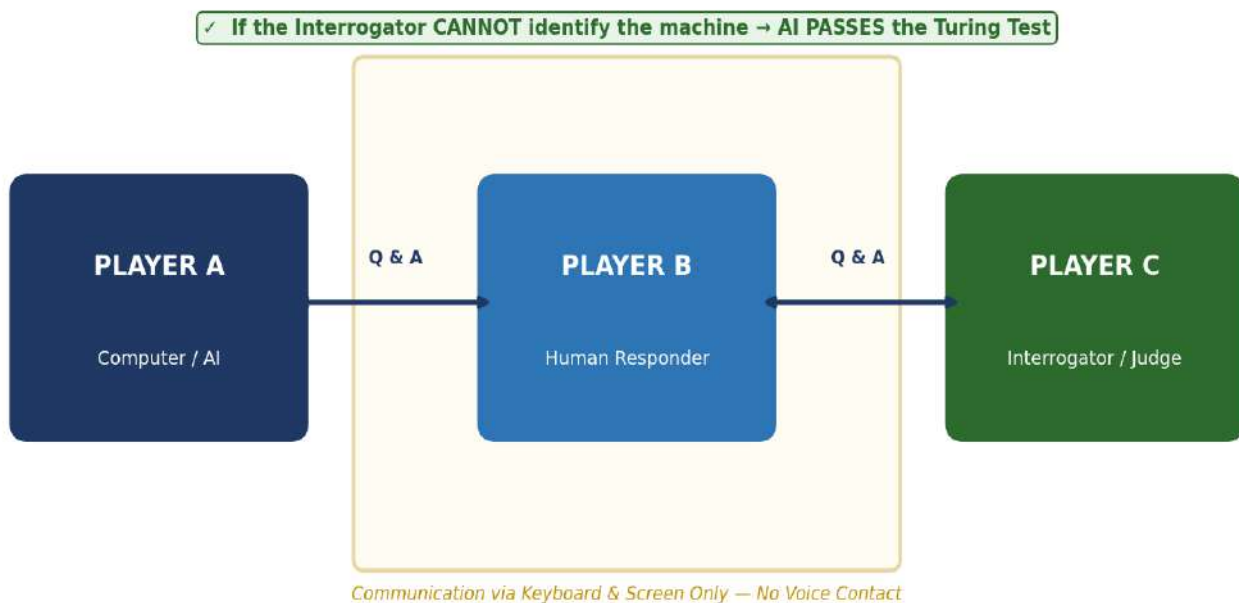


Figure 3: The Turing Test — Three-Player Imitation Game Setup

The Turing Test involves three players:

- **Player A — The Computer:** A machine attempting to appear human in its responses.
- **Player B — A Human Responder:** A human providing natural, human responses.
- **Player C — The Interrogator (Judge):** A human who communicates with both A and B and tries to determine which one is the machine.

The interrogator is aware that one of them is a machine but must identify which one based solely on questions and responses. The conversation is conducted via keyboard and screen only — so the result does not depend on the machine's ability to convert words into speech. The test result does not depend on each correct answer individually, but on how closely the machine's responses resemble human answers.

Sample exchange — Interrogator: "Are you a computer?"

Player A (Computer): "No, I am not." (then gives a slightly wrong answer to a maths question deliberately, to appear human.)

If the game ends and the interrogator cannot identify which player is the machine, the computer passes the test and is considered intelligent.

2.3 Purpose of the Turing Test

The purpose of the Turing Test is to provide a practical benchmark for determining whether a machine exhibits intelligent behaviour indistinguishable from that of a human. Rather than asking whether a machine "truly thinks," the Turing Test focuses on observable performance — if the machine behaves like a human, it can be treated as intelligent for practical purposes.

2.4 Major Criticism of the Turing Test

The major criticism is that the Turing Test only measures the ability to produce human-like textual responses and does not test for genuine understanding, consciousness, or reasoning. A machine could pass by clever pattern-matching without any real comprehension. This is illustrated by philosopher John Searle's Chinese Room thought experiment — a system can simulate intelligent responses without truly understanding anything.

Chapter 3

Intelligent Agents

3.1 What is an Intelligent Agent?

Figure 10: Intelligent Agent Cycle — Perceive → Process → Act

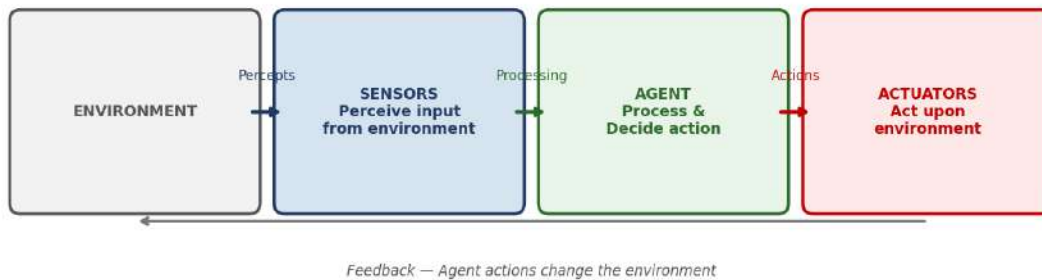


Figure 4: Intelligent Agent Cycle — Perceive → Process → Act

An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators. An AI system is composed of an agent and its environment. Artificial Intelligence is formally defined as a study of rational agents. A rational agent could be anything: a person, firm, machine, or software program.

An intelligent agent has the following characteristics:

- Collects input data from the environment through sensors such as cameras, microphones, and other sensing devices.
- Does real-time computation on the input and derives an appropriate action.
- Delivers its output to the environment through actuators such as screens, speakers, or motors.
- Has broad abilities including real-time problem solving, error-or-success rate analysis, and information processing.

3.2 Five Types of Agents

Figure 4: Five Types of Intelligent Agents — Key Features

Simple Reflex Agent	Model-Based Reflex Agent	Goal-Based Agent	Utility-Based Agent	Learning Agent
Current percept only	Internal state model	Knows goal state	Maximises utility	Learns from data
IF-THEN rules	Partial observability	Plans sequences	Handles trade-offs	4 key components
No memory	Tracks world	Searches for path	Best action chosen	Improves over time

Figure 5: Five Types of Intelligent Agents — Key Features

Type 1 — Simple Reflex Agent: Makes decisions on the basis of the current percept only and ignores the rest of the percept history. It implements a Condition-Action Rule (IF-THEN rule). For example: IF the floor is dirty, THEN vacuum the floor. It only succeeds when the environment is fully observable.

Example: A thermostat. When the room temperature drops below a set point, it turns on the heater. No memory of past temperatures — it reacts only to the current state.

Type 2 — Model-Based Reflex Agent: Works in a partially observable environment. It stores an internal state based on percept history and uses a model of how the world works. It tracks: (a) how the world evolves over time, and (b) how its own actions affect the world.

Example: A self-driving car that monitors other vehicles in adjacent lanes even when temporarily out of sensor range, using its internal model to make safe decisions.

Type 3 — Goal-Based Agent: Knows its current state and always needs to know the goal which describes desirable situations. It makes decisions by considering what action will bring it closer to its goal. It combines its model of the world with goal information.

Example: A taxi navigation agent that decides to turn left or right at a junction based on its destination (the goal).

Type 4 — Utility-Based Agent: Used when there are multiple possible solutions and the best possible outcome must be chosen. A utility function maps a state to a real number representing the degree of desirability. The agent selects the action that maximises its expected utility.

Example: An online auction bidding agent that selects the bid maximising its expected gain.

Type 5 — Learning Agent: Can learn from its past experiences and adapt automatically through learning. It has four components: (1) Learning Element — makes improvements from environment, (2) Critic Element — gives feedback on performance, (3) Performance Element — selects external actions, (4) Problem Generator — suggests new informative actions.

Example: A human learning to drive a car — gaining basic knowledge and improving through practice and feedback over time.

Chapter 4

The PEAS Framework

4.1 What is PEAS?

In AI, there are different rational agents used in various applications. PEAS is a framework used to systematically describe and classify intelligent agents. The PEAS framework ensures we fully define what an agent is trying to achieve, the world it operates in, how it acts on that world, and how it perceives it. PEAS stands for:

- **P** Performance Measure — what the agent is trying to achieve (its success criteria).
- **E** Environment — the world or situation the agent operates in.
- **A** Actuators — the tools or mechanisms through which the agent acts on the environment.
- **S** Sensors — the tools or mechanisms through which the agent perceives the environment.

The most intelligent agents are called rational agents. A rational agent considers all possibilities and chooses the best, most efficient action. PEAS provides a structured way to describe exactly what any intelligent agent is doing, in what environment, and with what tools.

4.2 Components of PEAS Explained in Detail

P — Performance Measure: The performance measure is the criterion that defines success for an agent. It answers the question: what do we want this agent to achieve? It varies entirely depending on the type of agent. A self-driving car is measured on safety, speed, and fuel efficiency. A medical diagnosis AI is measured on accuracy of diagnosis and minimising false results. A virtual assistant is measured on how correctly and quickly it understands and

completes user requests. The performance measure is set by the designer of the agent and acts as its goal.

E — Environment: The environment is the external world the agent lives and operates in. It is everything outside the agent itself. The environment provides sensory input to the agent and also receives the agent's actions as output. Environments can be very different in nature: a robot works in a physical factory environment; a smartphone assistant works in a digital environment of apps, data, and internet services; a self-driving car works in a complex, unpredictable road environment with weather, traffic, and pedestrians. The environment type determines which agent design is most appropriate.

A — Actuators: Actuators are the components through which an agent delivers its output — how it physically or digitally acts on its environment. Actuators are what the agent uses to change the state of the world around it. For a robot, actuators are its arms, wheels, grippers, and tools. For a self-driving car, actuators are the steering motor, accelerator, and brakes. For a smartphone assistant, actuators are the speaker (voice output), the screen (text/visual output), and the ability to send messages or open apps. Without actuators, an agent can perceive but cannot act.

S — Sensors: Sensors are the components through which an agent perceives and collects input data from its environment. Sensors are how the agent "sees" and "hears" the world. For a self-driving car, sensors include cameras, LIDAR, RADAR, and GPS. For a robot, sensors include cameras, force sensors, and proximity detectors. For a smartphone assistant, sensors include the microphone, camera, touchscreen, and GPS. The quality and accuracy of sensors directly determines how well an agent can understand its environment and make good decisions. Poor sensors lead to poor decisions, no matter how intelligent the agent.

4.3 PEAS Example 1: Automated Taxi Agent

Figure 5: PEAS Framework — Applied to an Automated Taxi Agent

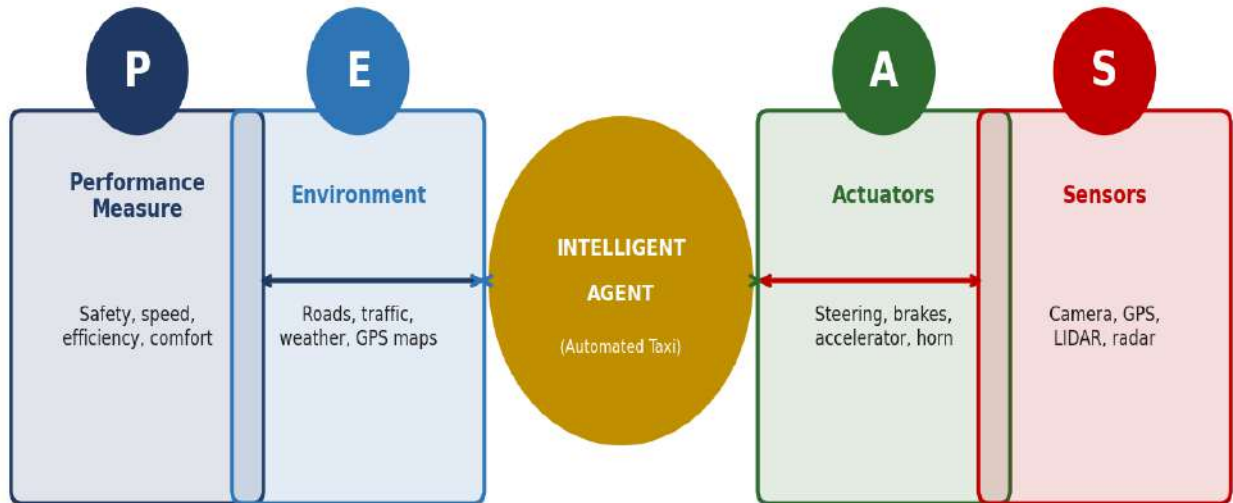


Figure 6: PEAS Framework — Automated Taxi Agent

PEAS Component	Description
P — Performance Measure	Safe driving, reach destination on time, obey traffic rules, maximise passenger comfort, minimise journey time and fuel consumption
E — Environment	Roads, highways, traffic lights, other vehicles, pedestrians, cyclists, weather conditions, GPS maps, construction zones
A — Actuators	Steering wheel motor, accelerator, brake pedal, gear shifter, horn, indicator lights, navigation display screen
S — Sensors	Front/rear/side cameras, GPS, speedometer, odometer, LIDAR/RADAR for obstacle detection, engine sensors, accelerometer

4.4 PEAS Example 2: Self-Driving Car Agent

A self-driving car is one of the most complex AI agents. Unlike a simple taxi agent, it must handle entirely unpredictable environments — other drivers, pedestrians, bad weather, and road hazards — all in real time.

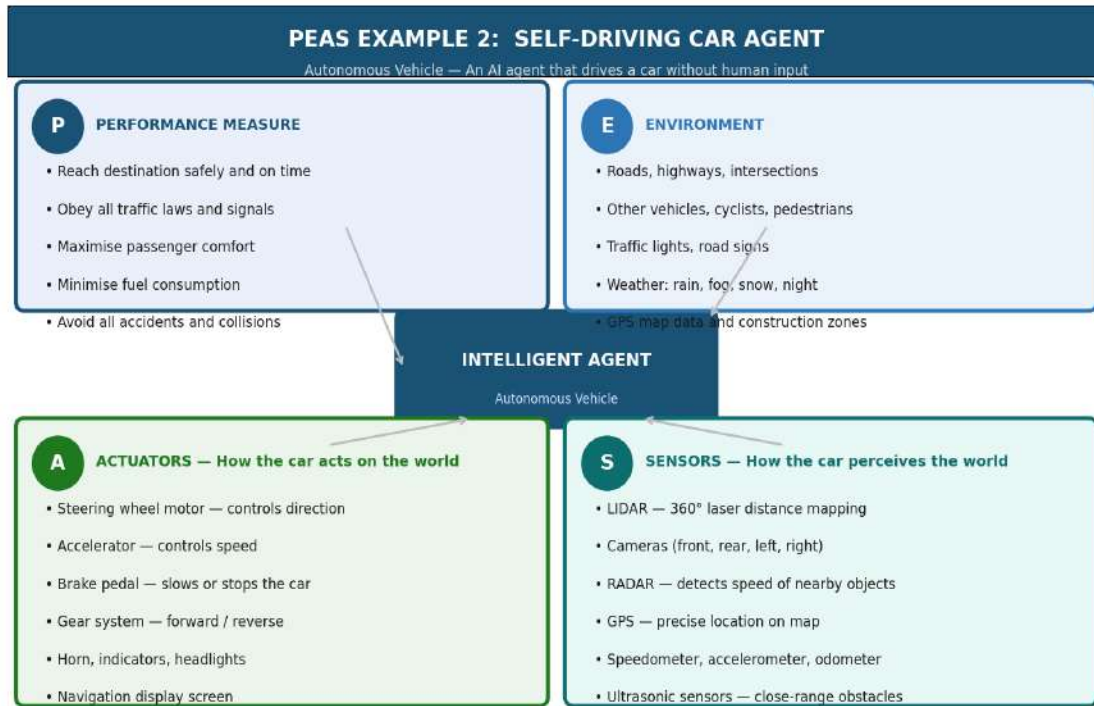


Figure 7: PEAS Framework — Self-Driving Car Agent

PEAS Component	Description
P — Performance Measure	Reach destination safely, obey all traffic laws, avoid all accidents, maximise comfort, minimise fuel use and travel time
E — Environment	Roads, highways, junctions, other vehicles, cyclists, pedestrians, traffic signals, weather (rain/fog/snow/night), GPS map data
A — Actuators	Steering wheel motor (direction), accelerator (speed), brake (stop), gear system (forward/reverse), horn, indicators, headlights, navigation screen
S — Sensors	LIDAR (360° laser mapping), cameras (all sides), RADAR (object speed), GPS (location), speedometer, accelerometer, ultrasonic sensors (close range)

Agent type: The self-driving car is a Model-Based Reflex Agent combined with Goal-Based and Utility-Based reasoning. It maintains an internal model of the road and surrounding traffic, uses goals (reach destination) and utility functions (safest + fastest route) to make decisions continuously in real time.

4.5 PEAS Example 3: Smartphone Virtual Assistant Agent

A smartphone virtual assistant (such as Siri or Google Assistant) is a software AI agent that understands human voice and text input and takes actions on behalf of the user.

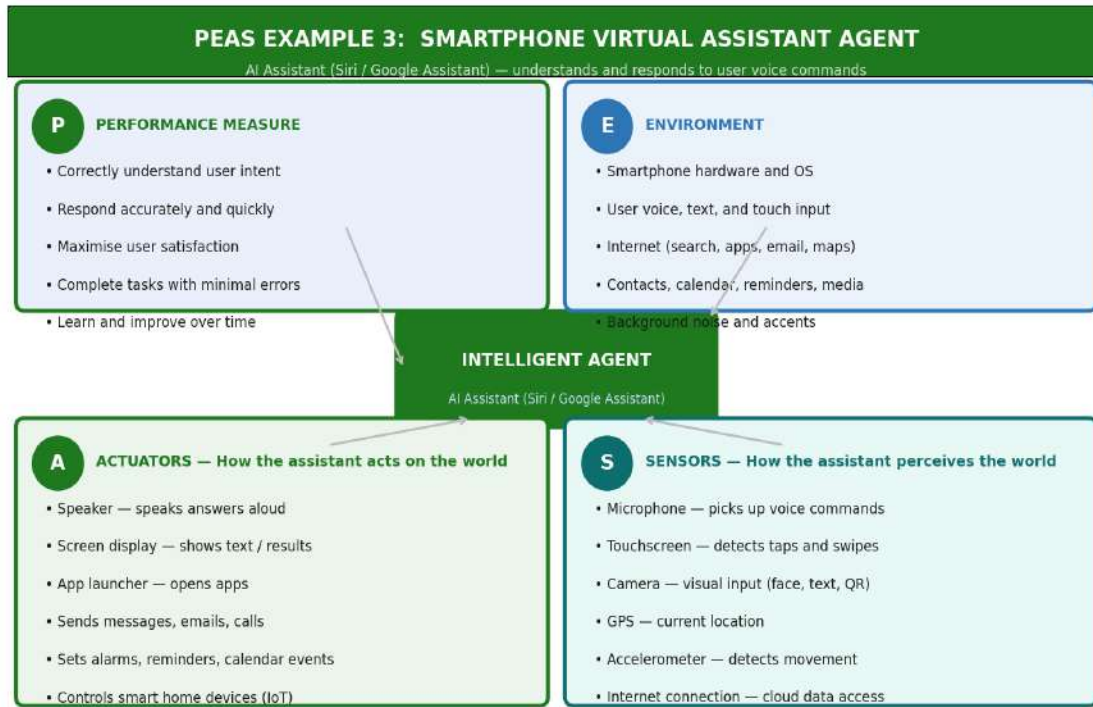


Figure 8: PEAS Framework — Smartphone Virtual Assistant Agent (Siri / Google Assistant)

PEAS Component	Description
P — Performance Measure	Correctly understand user intent, respond accurately and quickly, maximise user satisfaction, complete tasks with minimal errors, improve with use
E — Environment	Smartphone hardware, user voice/text/touch input, internet (search, apps, email, maps), contacts, calendar, media, background noise
A — Actuators	Speaker (spoken answers), screen display (text/visual), app launcher, message/call sender, alarm and reminder setter, smart home device controller (IoT)
S — Sensors	Microphone (voice commands), touchscreen (taps/swipes), camera (visual input, QR, face), GPS (location), accelerometer (movement), internet connection

Agent type: A smartphone assistant is a Learning Agent. It starts with baseline knowledge and continuously learns from the user's habits, preferences, voice patterns, and corrections to improve its accuracy and personalisation over time.

4.6 PEAS Example 4: Industrial Robot Agent

An industrial robot is a physical AI agent that operates in a factory or warehouse environment. It uses sensors to perceive its surroundings and actuators (arms, grippers, wheels) to physically interact with the world.

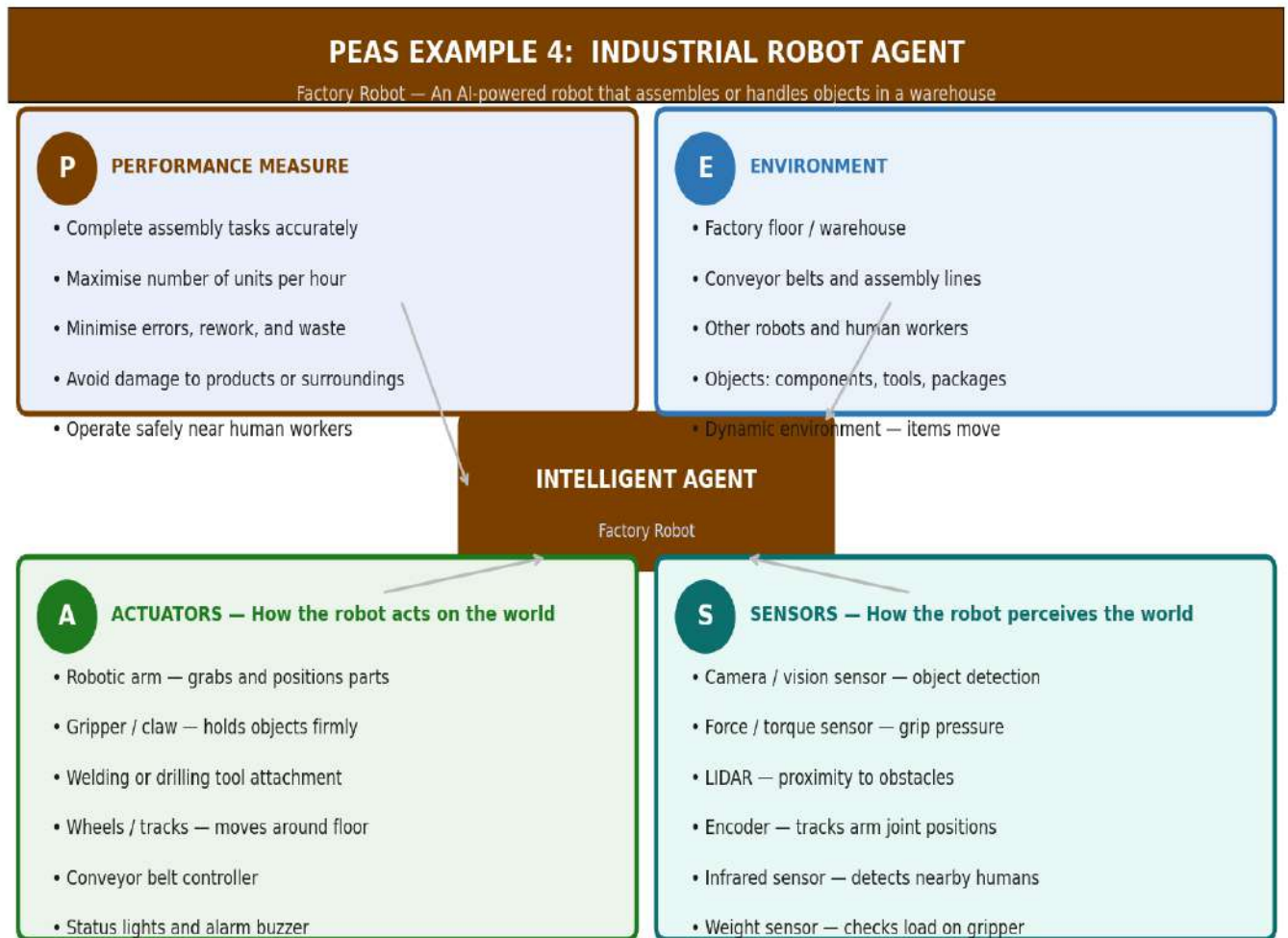


Figure 9: PEAS Framework — Industrial Factory Robot Agent

PEAS Component	Description
P — Performance Measure	Complete assembly tasks accurately, maximise units per hour, minimise errors and waste, avoid damage to products, operate safely near human workers
E — Environment	Factory floor or warehouse, conveyor belts, assembly lines, other robots and human workers, components, tools, packages, dynamic moving objects
A — Actuators	Robotic arm (grabs and positions parts), gripper/claw (holds objects), welding/drilling tool, wheels or tracks (movement), conveyor belt controller, status lights and alarm buzzer
S — Sensors	Camera/vision sensor (object detection), force/torque sensor (grip pressure), LIDAR (proximity to obstacles), encoder (arm joint position), infrared sensor (nearby humans), weight sensor (load on gripper)

Agent type: An industrial robot is typically a Goal-Based or Utility-Based Agent. It follows a defined sequence of goals (pick, place, weld) and may use utility functions to decide between multiple ways to complete a task — choosing the fastest, most precise, or safest method based on real-time sensor input.

4.7 Summary: PEAS Comparison Across All Four Examples

Agent	P — Goal	E — World	A — Acts Via	S — Perceives Via
Automated Taxi	Safe, timely arrival	Roads, traffic, weather	Steering, brake, accelerator	Cameras, GPS, LIDAR
Self-Driving Car	Safe drive, traffic laws	Roads, pedestrians, signals	Steering motor, brake, horn	LIDAR, RADAR, cameras, GPS
Phone Assistant	Correct, fast responses	Apps, voice, internet, calendar	Speaker, screen, app control	Microphone, camera, GPS, touch
Industrial Robot	Accurate assembly, speed	Factory floor, parts, workers	Arm, gripper, wheels, tools	Vision camera, force, LIDAR, IR

Chapter 5

Problem-Solving in Artificial Intelligence

5.1 What is Problem-Solving in AI?

Searching is the process of finding the required states or goals from a given set of states for a given problem. Search algorithms are one of the most important areas of Artificial Intelligence. The searching process is carried out by constructing a search tree, and search is regarded as a universal problem-solving technique.

In AI, problem-solving refers to AI techniques such as search algorithms and goal-oriented planning that can be used to systematically find a solution to a given problem. A planning agent is viewed as the problem generator with the solution being the plan. A problem-solving agent selects its actions based on its goal, with each action intended to reduce its distance from the goal.

5.2 Characteristics of a Well-Defined Problem

A well-defined problem in AI must have five key components:

- 1. Initial State:** The starting point or initial configuration of the problem. It is the state in which the agent begins. The initial state completely describes the situation before any action is taken.
- 2. Actions (Operators):** A description of the set of possible actions available to the agent in any given state. These are the operators that transition the agent from one state to another. For example, in a chess game, the available actions are all the legal moves a piece can make.
- 3. Transition Model (Result Function):** A description of what each action does — it specifies which new state results from performing a given action in a given state. Together, the initial state, actions, and transition model define the state space of the problem.

4. Goal Test: A function that determines whether the current state is the goal state — that is, when the problem has been successfully solved. In some problems there is a single goal state; in others, any state satisfying a set of conditions is acceptable.

5. Path Cost: A function that assigns a numeric cost to each path taken from the initial state to the current state. It is used to evaluate the quality or efficiency of a solution. We aim to minimise path cost — for example, the shortest distance or fastest time.

5.3 Example: Route-Finding Problem

The route-finding problem is a classic well-defined AI problem. Consider navigating from Kathmandu to Pokhara:

- **Initial State:** The agent is currently located in Kathmandu.
- **Actions:** Drive to any directly connected neighbouring city (e.g., Kathmandu to Naubise).
- **Transition Model:** Performing "drive to Mugling" while in Naubise results in the new state of being in Mugling.
- **Goal Test:** Is the agent currently in Pokhara? If yes, the problem is solved.
- **Path Cost:** Total distance in kilometres or total travel time from Kathmandu to current location.

This is a classic well-defined AI search problem where algorithms such as BFS, DFS, Uniform Cost Search, or A* can be applied to find the optimal or shortest path.

Chapter 6

Search Algorithms

6.1 Uninformed Search vs. Informed Search

Figure 7: Uninformed vs Informed Search — Relative Efficiency

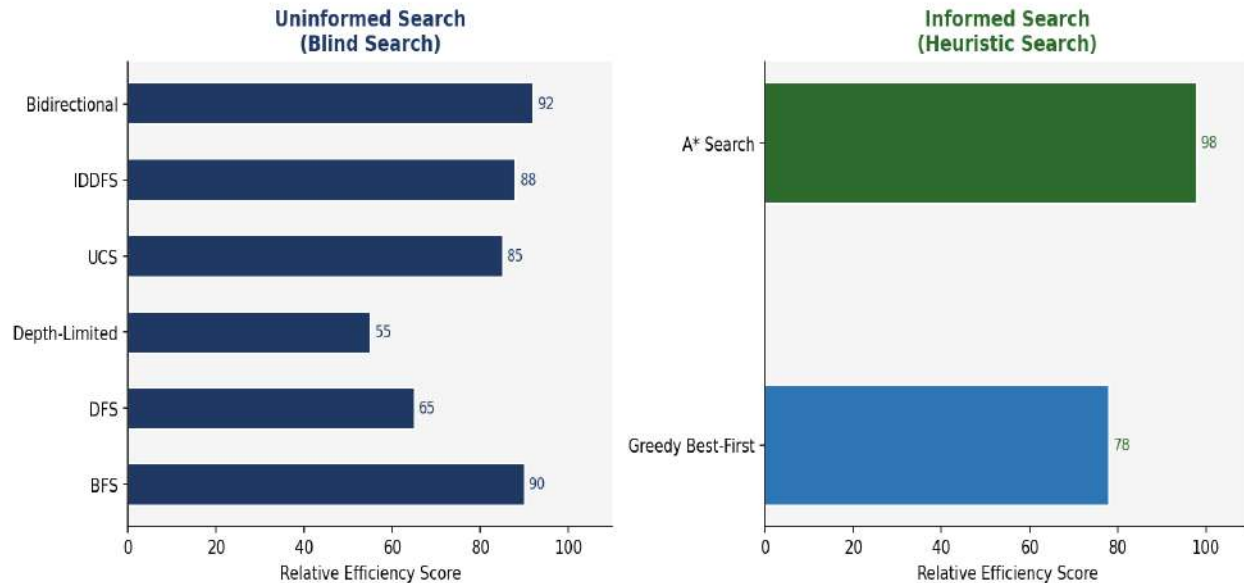


Figure 10: Uninformed vs. Informed Search — Relative Efficiency Comparison

Feature	Uninformed Search (Blind)	Informed Search (Heuristic)
Domain Knowledge	None — no extra information about the problem	Uses heuristic function $h(n)$ as domain knowledge
Strategy	Brute-force — explores all possible paths systematically	Prioritises paths that appear closer to the goal
Closeness to Goal	No idea how close the goal is	Estimates closeness via the heuristic $h(n)$
Efficiency	Less efficient — may explore many unnecessary nodes	More efficient — explores fewer nodes
Common Examples	BFS, DFS, UCS, IDDFS, Bidirectional	Best-First Search, Greedy, A* Search
Also Called	Blind search algorithms	Heuristic search algorithms

6.2 Breadth-First Search (BFS)

Figure 6: Breadth-First Search — Level-by-Level Tree Traversal

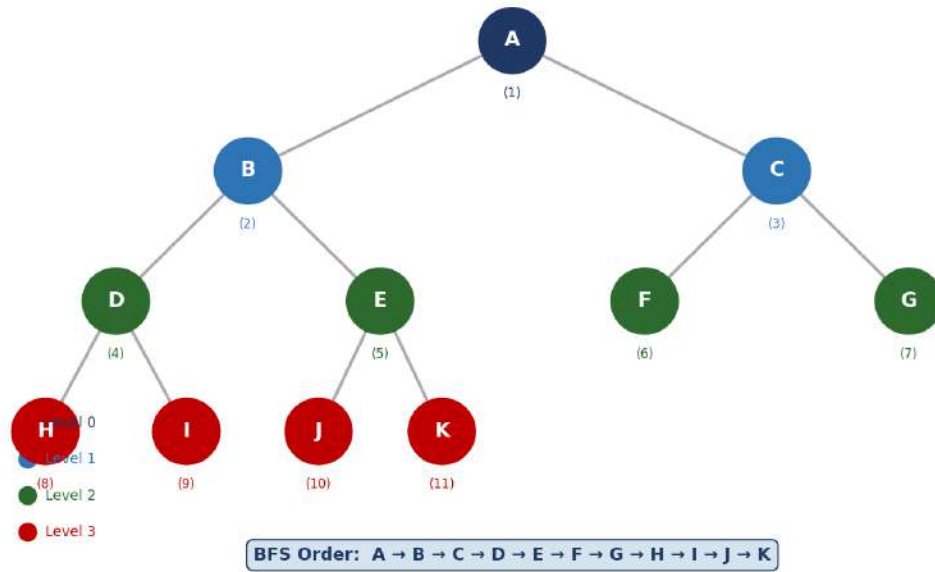


Figure 11: Breadth-First Search (BFS) — Level-by-Level Tree Traversal

Breadth-First Search (BFS) is the most common search algorithm used for traversing a tree or graph. BFS searches breadth-wise — it starts at the root node and expands all successor nodes at the current level before moving to nodes of the next level. It uses a FIFO (First In, First Out) queue data structure.

BFS Traversal Order: A → B → C → D → E → F → G → H → I → J → K (level by level, as shown in Figure 6 above).

Property	Value	Explanation
Completeness	Yes — Complete	Always finds a solution if one exists at any finite depth.
Optimality	Yes — Optimal	Always finds the shallowest (minimum-step) solution first.
Time Complexity	$O(b^d)$	b = branching factor, d = depth of shallowest solution.
Space Complexity	$O(b^d)$	All nodes at each level must be stored in memory.

Advantage: BFS will always provide a solution if any solution exists. If there are more than one solution for a given problem, BFS will provide the minimum (optimal) solution which requires the least number of steps.

Disadvantage: BFS requires a very large amount of memory since each level of the tree must be saved into memory before the next level can be explored. It is also slow if the solution is very deep in the tree, making it impractical for large or deep search spaces.

6.3 Other Uninformed Search Algorithms

Depth-First Search (DFS): Starts from the root node and follows each path to its deepest node before backtracking. Uses a LIFO stack data structure. Requires less memory than BFS but is not complete (can get stuck in loops) and is not guaranteed to find the shortest path.

Depth-Limited Search (DLS): Similar to DFS but with a fixed depth limit — the algorithm will not explore nodes beyond the specified depth. Has two failure conditions: standard failure (no solution exists) and cutoff failure (no solution within the depth limit).

Uniform Cost Search (UCS): Finds the path with the lowest total cumulative cost from the root to the goal. At each step, UCS expands the node with the lowest path cost. It is complete and optimal when all step costs are positive.

Bidirectional Search: Runs two simultaneous searches — one forward from the initial state and one backward from the goal state. They meet in the middle, reducing time complexity from $O(b^d)$ to $O(b^{(d/2)})$.

Iterative Deepening DFS (IDDFS): Combines the memory efficiency of DFS with the completeness and optimality of BFS by gradually increasing the depth limit. Considered the

best uninformed search algorithm when the search space is large and depth of the goal is unknown.

6.4 Informed Search and A* Search

Figure 9: A* Search — Evaluation Function $f(n) = g(n) + h(n)$

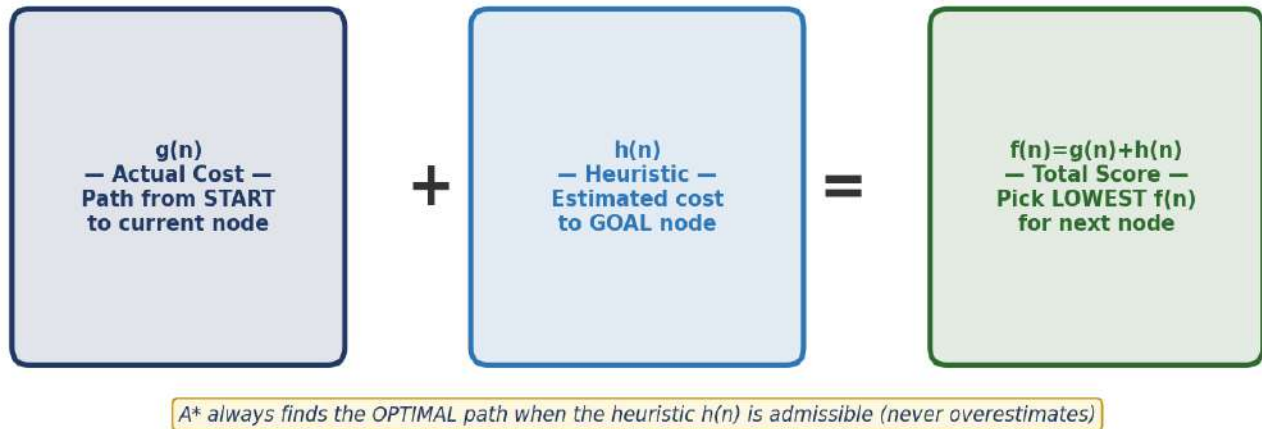


Figure 12: A* Search — Evaluation Function $f(n) = g(n) + h(n)$

Heuristic Function $h(n)$: A heuristic function estimates how close a state n is to the goal. It must be admissible — it should never overestimate the true cost. The heuristic guides the search towards the goal more efficiently than uninformed methods.

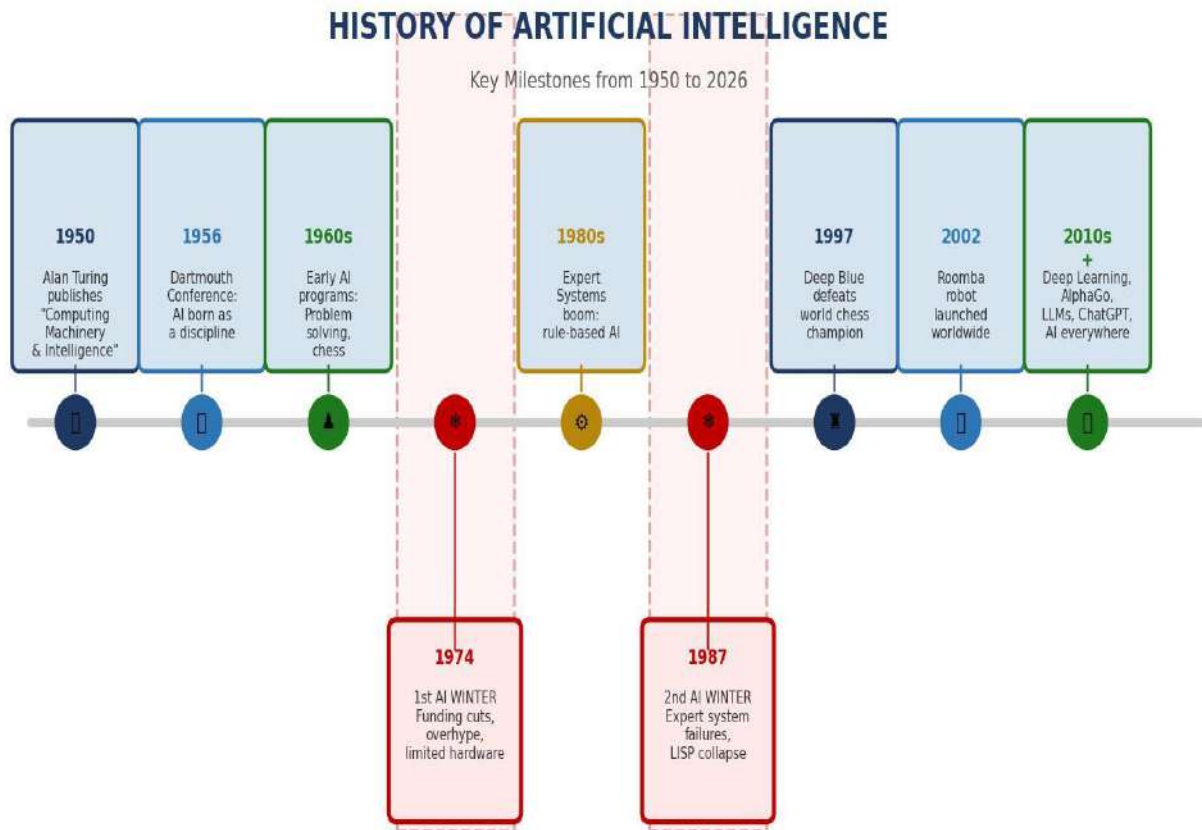
Best-First Search: Uses the heuristic function $h(n)$ to select the node that appears closest to the goal at each step. Evaluation function: $f(n) = h(n)$. It can switch between BFS and DFS, but is not guaranteed to be optimal.

A* Search: The most widely used informed search algorithm. It combines the actual cost from the start node $g(n)$ and the estimated cost to the goal $h(n)$. Evaluation function: $f(n) = g(n) + h(n)$. A* is both complete and optimal when the heuristic $h(n)$ is admissible. It always finds the shortest path.

Chapter 7

History of Artificial Intelligence

AI has evolved over more than 70 years — from a bold idea in the 1950s to a technology transforming every industry today. The complete journey is shown in the timeline below.



OW.

Figure 13: History of AI — Complete Milestones Timeline 1950–2026

7.1 The Two Founding Figures

Two people are most responsible for creating AI as a field of study:

TWO FOUNDERS OF ARTIFICIAL INTELLIGENCE



Figure 14: Alan Turing (1950) and John McCarthy (1956) — The Two Founders of AI

Alan Turing (1950): Proposed the Turing Test, asking "Can machines think?" His paper established the philosophical foundation of AI and defined how we measure machine intelligence.

John McCarthy (1956): Coined the term "Artificial Intelligence" and organised the Dartmouth Conference — officially founding AI as an academic discipline. He also invented the LISP programming language (1958).

7.2 AI Winter — Causes and Revival

Twice in history, AI crashed into an "AI Winter" — periods of lost funding and interest. The diagram below shows exactly why it happened and what brought AI back:

AI WINTER — Causes and Revival

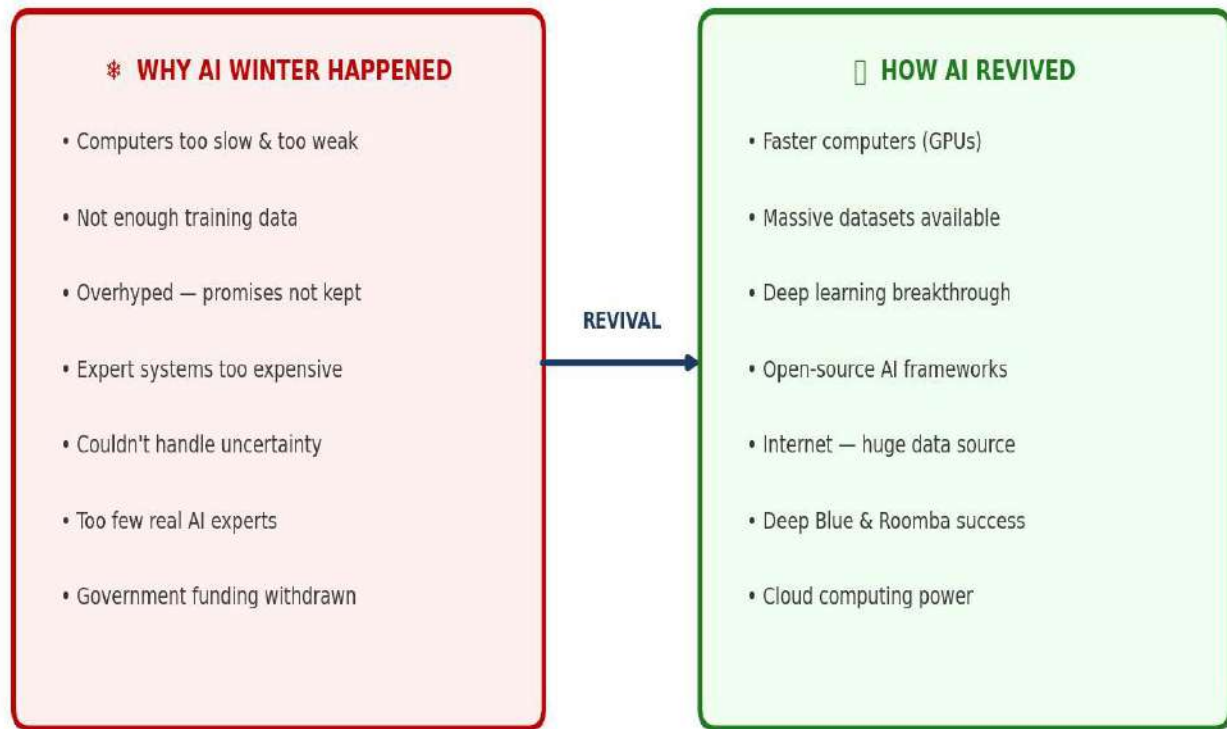


Figure 15: AI Winter — Why It Happened (left) and What Revived AI (right)

First Winter (1974–1980): Computers were too slow, data was scarce, and AI failed to meet overhyped promises. Government funding was cut worldwide.

Second Winter (1987–1993): Expensive expert systems failed commercially. They could not learn, adapt, or handle uncertainty. The LISP machine market collapsed and many AI companies shut down.

7.3 Modern AI — The Breakthrough Era

Faster hardware, massive datasets, and deep learning triggered a powerful revival. The milestones below define the modern age of AI:

MODERN AI – Key Milestones (1997 - 2026)

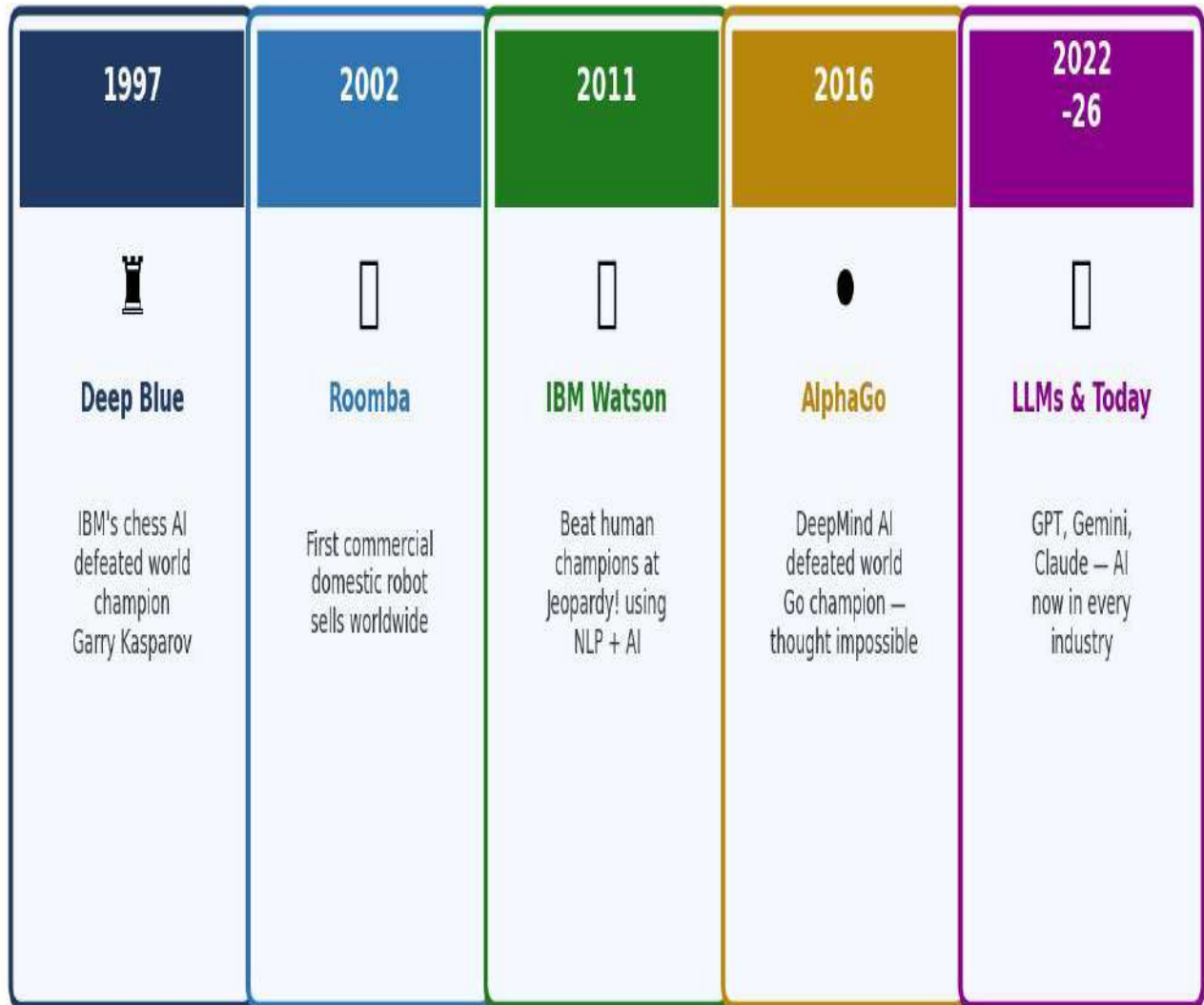


Figure 16: Modern AI Milestones — From Deep Blue (1997) to Large Language Models (2022–2026)

From Deep Blue's chess victory in 1997 to today's Large Language Models, AI has gone from laboratory experiments to being embedded in smartphones, hospitals, vehicles, and homes across the entire world.

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Appendix A: Glossary of Key Terms

The table below explains the important terms used throughout this report in simple language.

Term	Simple Meaning
Artificial Intelligence (AI)	The simulation of human intelligence in machines, enabling them to learn, reason, and solve problems.
Rational Agent	Any entity (machine, person, or software) that perceives its environment and acts to maximise its performance measure.
Percept	The agent's input at any given moment — what it currently senses from the environment.
Actuator	The mechanism through which an agent acts on its environment (e.g., motors, screens, speakers).
Sensor	The mechanism through which an agent perceives its environment (e.g., cameras, microphones, GPS).
PEAS	Performance Measure, Environment, Actuators, Sensors — a framework for describing intelligent agent systems.
Turing Test	A test to evaluate whether a machine can exhibit intelligent behaviour indistinguishable from a human.
Heuristic Function $h(n)$	An estimate of how close a state n is to the goal, used by informed search algorithms.
BFS	Breadth-First Search — explores all nodes at the current level before moving to the next; guarantees the shortest path.
DFS	Depth-First Search — explores as far as possible along each branch before backtracking.
A* Search	An informed search algorithm combining actual path cost $g(n)$ and heuristic estimate $h(n)$ to find the optimal path.
AI Winter	A period of reduced funding and interest in AI research due to failed promises and unmet expectations.
Fully Observable	An environment where sensors give access to the complete state at all times.
Deterministic	An environment where the next state is completely determined by the current state and action taken.
Stochastic	An environment where the next state involves some element of randomness or uncertainty.
State Space	The set of all possible states reachable from the initial state through any sequence of actions.
Goal Test	A function that checks whether the current state satisfies the conditions of the goal state.

Path Cost	The total cost of the sequence of actions taken from the initial state to the current state.
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Appendix B: Types of Agents — Full Comparison

The table below provides a complete side-by-side comparison of all five types of intelligent agents for easy revision.

Agent Type	Memory?	Goals?	Utility?	Learns?	Example
Simple Reflex Agent	No	No	No	No	Thermostat, automatic door sensor
Model-Based Reflex Agent	Yes — internal state	No	No	No	Self-driving car tracking lanes
Goal-Based Agent	Yes	Yes	No	No	Taxi navigation agent
Utility-Based Agent	Yes	Yes	Yes	No	Chess agent, auction bidder
Learning Agent	Yes	Yes	Yes	Yes	Spam filter, voice assistant

Appendix C: Search Algorithms — Full Comparison

The table below summarises all major search algorithms covered in this report.

Algorithm	Type	Complete?	Optimal?	Time	Space	Heuristic?
BFS	Uninformed	Yes	Yes	$O(b^d)$	$O(b^d)$	No
DFS	Uninformed	No	No	$O(b^m)$	$O(bm)$	No
Depth-Limited	Uninformed	No	No	$O(b^l)$	$O(bl)$	No
UCS	Uninformed	Yes	Yes	$O(b^{(C/e)})$	Same	No
IDDFS	Uninformed	Yes	Yes	$O(b^d)$	$O(bd)$	No
Bidirectional	Uninformed	Yes	Yes	$O(b^{(d/2)})$	Same	No
Greedy Best-First	Informed	No	No	$O(b^m)$	$O(b^m)$	Yes — $h(n)$
A* Search	Informed	Yes	Yes	$O(b^d)$	$O(b^d)$	Yes — $f(n)=g(n)+h(n)$